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MEMOL: 通过多头注意力进行多模态学习的专家混合, 用于预测药物毒性

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摘要

背景 和 **Objective**: toxicity prediction is essential in drug development but several challenges remain due to the high complexity of molecular mechanisms. In this regard, toxicity prediction methods using deep learning have emerged as promising solutions.

然而, 由于分子机制的高复杂性, 毒性预测仍然需要进一步探索。在这项研究中, 我们提出 MEMOL: 混合专家的多模态学习通过多头注意力。这个新颖的框架集成了分子图像、图形和指纹的多种模态。此外, MEMOL 增强了特征提取和模态融合通过自-和交叉-注意力机制, 导致预测性能和效率的提高。

方法: 在这项研究中, 我们提出 MEMOL: 混合专家的多模态学习通过多头注意力。这个新颖的框架集成了分子图像、图形和指纹的多种模态。此外, MEMOL 增强了特征提取和模态融合通过自-和交叉-注意力机制, 导致预测性能和效率的提高。

结果: 在一个毒性基准数据集上的比较实验表明, MEMOL 优于第二最佳模型。此外, 多模态学习分析表明, 所有三种模态的使用都导致了性能改进。

结论: MEMOL 展示了在毒性预测中的优越性能, 实现了与跨多个数据集的一致改进, 并为集成分子表示提供了一个鲁棒的框架。

结论: MEMOL demonstrates superior performance in toxicity prediction, achieving consistent improvement across diverse datasets and offering a robust framework for integrating molecular representations. 相关

1. 介绍

药物开发过程高度结构化且复杂, 涉及广泛的研究、严格的测试以及跨多个领域的协作。美国食品药品监督管理局 (FDA) 将这一复杂过程划分为五个关键阶段: 发现与开发、临床前研究、临床研究、FDA 审查以及上市后安全监测。

在发现和开发阶段, 研究人员识别针对特定疾病机制的候选化合物, 并借助高通量筛选和计算机辅助药物设计 [2]。然后, 对有前景的候选药物进行代谢、剂量和安全性评估, 之后进入临床前研究阶段, 在实验室和动物实验中进一步评估生物活性和毒性 [3]。随后进行临床研究, 包括分阶段的人体试验: 早期阶段侧重安全性和剂量, 而后期阶段测试疗效并监测不良反应。如果成功, 将提交新药申请以供 FDA 审查, 审查内容包括试验数据 and 生产质量。获批后, 药物进入上市后的安全性监测阶段。

追踪长期风险 [4]。总体而言, 药物开发通常需要 10–15 年, 费用为 10–30 亿美元 [5]。

尤其是, 毒性评估是整个药物开发过程中的重要环节, 在确保患者安全方面起着至关重要的作用。然而, 传统的毒性评估方法面临诸多挑战。这些研究成本高昂, 并且在动物实验方面引发伦理问题。为了解决这些限制, 人工智能 (AI) 已成为分析复杂数据集和提高预测准确性的有力工具。尤其是, 深度学习方法显著提升了毒性预测模型的性能, 各种架构已成功应用于该领域。

例如, 基于 Transformer 的模型通过处理与药物相关的序列 [11,12] 或图像数据 [13] 来进行毒性预测。通过使用 Transformer 编码器将分子结构式或临床试验数据编码为序列表示, 这些模型能够有效预测各种毒性。视觉 Transformer

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MEMOL: Mixture of experts for multimodal learning through multi-head attention to predict drug toxicity

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ABSTRACT

Background and Objective: Accurate toxicity prediction is essential in drug development, but several challenges remain due to the high complexity of molecular mechanisms. In this regard, toxicity prediction methods using deep learning have emerged as promising solutions, with multimodal approaches by integrating diverse molecular modalities. However, an effective fusion of multiple modalities to achieve precise toxicity predictions still requires further exploration.

Methods: In this study, we propose MEMOL, mixture of experts for multimodal learning through multi-head attention. This novel framework integrates modalities of molecular images, graphs, and fingerprints, and applies a sparse mixture of experts directly into the attention mechanisms. In addition, MEMOL enhances feature extraction and modality fusion through self- and cross-attention mechanisms, leading to improved predictive performance and efficiency.

Results: A comparative experiment on several toxicity benchmark datasets demonstrated that MEMOL outperformed state-of-the-art methods, achieving up to 8.33% higher AUROC and 9.11% higher AUPRC than the second-best model. Furthermore, the multimodal learning analysis showed that the employment of all three modalities consistently resulted in considerable performance improvements, and the ablation study revealed that a sparse mixture of experts enabled effective multimodal integration for drug toxicity prediction.

Conclusion: MEMOL demonstrates superior performance in toxicity prediction, achieving consistent improvements across diverse datasets and offering a robust framework for integrating molecular representations.

1. Introduction

The drug development process is highly structured and complex, involving extensive research, rigorous testing, and collaboration across various fields. The U.S. Food and Drug Administration (FDA) classifies this complex process into five key stages: discovery and development, preclinical research, clinical research, FDA review, and post-market safety monitoring [1].

During the discovery and development phase, researchers identify candidate compounds targeting specific disease mechanisms, aided by high-throughput screening and computer-aided drug design [2]. Promising candidates are then evaluated for metabolism, dosage, and safety before progressing to preclinical studies, where laboratory and animal experiments further assess biological activity and toxicity [3]. Clinical research follows with phased human trials: early stages focus on safety and dosage, while later phases test efficacy and monitor adverse reactions. If successful, a New Drug Application is submitted for FDA review, where trial data and manufacturing quality are examined. Upon approval, the drug enters post-market safety monitoring to

track long-term risks [4]. Overall, drug development typically requires 10–15 years and costs \$1–3 billion [5].

In particular, toxicity assessment is an essential part of the entire drug development process, playing a crucial role in ensuring safety for patients. However, traditional toxicity evaluation methods face several challenges. These studies are costly, and raise ethical concerns regarding animal testing. To address these limitations, artificial intelligence (AI) has emerged as a powerful tool for analyzing complex datasets and improving predictive accuracy [6,7]. Especially, deep learning approaches have significantly enhanced the performance of toxicity prediction models, with various architectures successfully applied to this domain [8–10].

For example, Transformer-based models are employed in toxicity prediction by processing drug-related sequences [11,12] or image data [13]. By encoding molecular structural formulas or clinical trial data as sequential representations using a Transformer encoder, these models effectively predict various toxicities. Vision Transformer

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(ViT) [14], 特别将分子结构图像分割成小块, 将其转换为序列表示, 并分析块间关系。这种方法使模型能够捕捉空间和视觉模式, 从而有助于精确识别分子中的关键结构特征。在另一个例子中, 图神经网络 (GNNs) 被应用于分子图, 其中原子对应于节点, 化学键对应于边。通过利用图中的关系信息, GNNs能够有效捕捉原子间的相互作用和与毒性预测相关的整体结构特性 [15,16]。

多模态学习 [17] 整合了多种分子数据源, 以提供对药物特性全面的理解。通过融合来自多种分子表示的互补信息, 这些方法能够捕捉在仅依赖单一模态时可能被忽略的结构特征。例如, 分子图像保留空间排列信息, 分子图编码原子间相互作用, 而指纹 (FP) 则以紧凑向量总结化学子结构。利用这种异构输入已在药物毒性研究中显示出改进的预测性能 [13,18,19]。然而, 由于数据结构和表示的根本差异, 有效整合多模态仍具有挑战性。

为应对这些挑战, 我们提出了一种名为 MEMOL 的新型框架, 该框架是基于 MoE 的多模态学习, 并结合了多头注意力机制。与之前的多模态模型 [13,18–20], 不同, MEMOL 直接在注意力机制中引入了稀疏专家混合 (MoE) [21–25] [26]。在多头自注意力 (MSA) 和多头交叉注意力 (MCA) [27], 中, MoE 会在计算注意力之前, 通过专门的专家投影对特定模态的表示进行选择性地路由。该架构使模型能够动态地强调每种模态及其交互中最有信息量的表示, 超越了静态融合方案。通过将专家的专门化与注意力过程对齐, MEMOL 在毒性预测方面建立了新的范式, 同时提升了模型的表现力和灵活性。

实验结果表明, MEMOL在药物毒性预测中达到了最先进的性能, 在 AUROC上比第二好的模型高出8.33%, 在AUPRC上高出9.11%。这些发现验证了将稀疏MoE整合到注意力机制中的有效性, 突出了不同模态的互补作用以及专家引导的多模态注意力的新颖性。

2. 初步事项

2.1. 多头注意力

注意力机制 [27] 对元素之间的成对依赖关系进行建模, 从而能够捕捉上下文关系。其数学定义如下:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (1)$$

其中 Q 、 K 、 V 分别是查询矩阵、键矩阵和值矩阵, 而 $\sqrt{d_k}$ 是用于数值稳定性的缩放因子。

自注意力使用 $Q = XW_Q$ 、 $K = XW_K$ 、 $V = XW_V$, 而交叉注意力则应用来自不同来源的 $Q = X_1W_Q$ 、 $K = X_2W_K$ 、 $V = X_2W_V$ 。两者遵循相同的公式, 但在 Q 、 K 、 V 的来源上有所不同。

多头注意力在不同的投影下并行执行 H 次注意力计算, 并将结果连接起来:

$$\text{Multi_Head}(Q, K, V) = \text{Concat}(A_1, A_2, \dots, A_H), \quad (2)$$

$$A^h = \text{Attention}(Q^h, K^h, V^h), \quad h = 1, 2, \dots, H. \quad (3)$$

这里 A^h 是第 h 个头的输出, 根据公式 (1) 用其自身的参数计算。所有头的拼接为下游任务提供了更丰富的表示。

2.2. 视觉变换器

ViT 将 Transformer 架构适应于图像数据。输入图像被分割为不重叠的 $P \times P$ 个小块, 每个小块被展平为一个令牌。为了保留空间顺序, 会加入位置嵌入, 并在最前面添加一个类别令牌, 以便模型能够输出全局表示。然后, 该序列通过一组 Transformer 编码器层进行处理。

每个编码器层都应用多头自注意力 (MSA) 和带残差连接的多层感知器 (MLP) 以及层归一化。前向传播表示为:

$$z'_l = \text{MSA}(\text{LN}(z_{l-1})) + z_{l-1}, \quad l = 1, \dots, L, \quad (4)$$

$$z_l = \text{MLP}(\text{LN}(z'_l)) + z'_l, \quad l = 1, \dots, L. \quad (5)$$

在公式(4)中, 输入 z_{l-1} 被归一化并传入多头自注意力 (MSA), 随后通过残差连接得到 z'_l 。公式(5)展示了第二次归一化, 之后经过多层感知机 (MLP) 和另一残差加法, 得到最终的 z_l 。通过堆叠 L 层, ViT 能够逐步将局部图像块信息整合为全局表示。

2.3. 图神经网络

图神经网络 (GNNs) 旨在处理图结构数据, 其中节点表示实体, 边表示它们之间的关系。通过迭代传播信息, GNN 能够学习既反映局部邻域又反映全局结构的节点表示。在代表性模型中, 图注意力网络 (GAT) [29] 和图同构网络 (GIN) [30] 被广泛研究。

GAT通过引入注意力机制自适应地为邻居加权, 从而扩展了图卷积网络 [31]。节点 i 的嵌入更新如下:

$$h'_i = \sigma\left(\sum_{j \in N(i)} \alpha_{ij} W h_j\right), \quad (6)$$

其中 h_j 是邻居 j 的特征, W 是可学习的权重矩阵, α_{ij} 是归一化的注意力分数。这使模型在聚合特征时能够更加重视信息量更大的邻居。

GIN 侧重于区分图结构的表达能力。节点 i 的嵌入公式如下:

$$h'_i = \text{MLP}\left((1 + \epsilon) h_i + \sum_{j \in N(i)} h_j\right), \quad (7)$$

其中 ϵ 是一个可学习的参数, 用于控制自信息和邻居聚合之间的平衡。求和操作确保了排列不变性, 而多层感知机 (MLP) 提高了表达能力, 使得 GIN 在理论上与 Weisfeiler–Lehman 测试一样强大。

2.4. 分子指纹

分子指纹 (FPs) 将分子结构转换为固定长度的向量, 用于相似性搜索、性质预测和筛选。通过以紧凑的形式编码化学信息, 分子指纹在化学信息学和药物发现中被广泛使用。

功能类指纹 (FCFP) [32], 由 Morgan 算法 [33], 得出, 通过迭代地结合邻近原子和键来优化原子标签。这既捕捉了结构细节, 也涵盖了功能角色, 如键类型或供体/受体基团。

药效基团二维 (Pharm2D) [34] 强调影响分子活性的药理学特征。它在二维框架中表示氢键供体/受体、疏水区域和带电基团等模式。通过关注这些功能特性, Pharm2D 提供了与基于结构的指纹互补的视角。

(ViT) [14], in particular, segments molecular structure images into small patches, converts them into sequential representations, and analyzes inter-patch relationships. This method enables the model to capture spatial and visual patterns, facilitating precise identification of key structural features within molecules. In another example, Graph Neural Networks (GNNs) are applied to molecular graphs, where atoms correspond to nodes and chemical bonds to edges. By leveraging relational information within these graphs, GNNs effectively capture atomic interactions and overall structural properties relevant to toxicity prediction [15,16].

Multimodal learning [17] integrates diverse molecular data sources to provide a comprehensive understanding of drug properties. By fusing complementary information from multiple molecular representations, these approaches capture structural features that may be overlooked when relying on a single modality. For example, molecular images preserve spatial arrangements, molecular graphs encode atomic interactions, and fingerprints (FP) summarize chemical substructures in compact vectors. Leveraging such heterogeneous inputs has shown improved predictive performance in drug toxicity studies [13,18,19]. However, effectively integrating modalities remains challenging due to fundamental differences in data structures and representations.

To address these challenges, we propose a novel framework called MEMOL which is MoE-based multimodal learning with multi-head attention. Unlike previous multimodal models [13,18–20], MEMOL incorporates a sparse Mixture of Experts (MoE) [21–25] directly into the attention mechanisms [26]. In both Multi-head Self-Attention (MSA) and Multi-head Cross-Attention (MCA) [27], the MoE selectively routes modality-specific representations through specialized expert projections before computing attention. This architecture enables the model to dynamically emphasize the most informative representations from each modality and their interactions, going beyond static fusion schemes. By aligning expert specialization with the attention process, MEMOL establishes a new paradigm in toxicity prediction, enhancing both expressiveness and flexibility.

The experimental results demonstrated that MEMOL achieved state-of-the-art performance in drug toxicity prediction, outperforming the second-best model by 8.33% in AUROC and 9.11% in AUPRC. These findings validate the effectiveness of integrating sparse MoE into attention, highlighting both the complementary roles of different modalities and the novelty of expert-guided multimodal attention.

2. Preliminaries

2.1. Multi-head attention

The attention mechanism [27] models pairwise dependencies between elements, enabling the capture of contextual relationships. It is mathematically defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (1)$$

where Q, K, V are query, key, and value matrices, and $\sqrt{d_k}$ is a scaling factor for numerical stability.

Self-attention uses $Q = XW_Q$ 、 $K = XW_K$ 、 $V = XW_V$, while cross-attention applies $Q = X_1W_Q$ 、 $K = X_2W_K$ 、 $V = X_2W_V$ from distinct sources. Both follow the same formulation but differ in the origin of Q, K, V .

Multi-head attention performs attention H times in parallel with different projections and concatenates the results:

$$\text{Multi_Head}(Q, K, V) = \text{Concat}(A_1, A_2, \dots, A_H), \quad (2)$$

$$A^h = \text{Attention}(Q^h, K^h, V^h), \quad h = 1, 2, \dots, H. \quad (3)$$

Here A^h is the output of the h th head, computed as in Eq. (1) with its own parameters. Concatenation of all heads yields a richer representation for downstream tasks.

2.2. Vision transformer

ViT adapts the Transformer architecture for image data. An input image is divided into non-overlapping $P \times P$ patches, each flattened into a token. Positional embeddings are added to preserve spatial order, and a class token is prepended so the model can output a global representation. This sequence is then processed by a stack of Transformer encoder layers.

Each encoder layer applies multi-head self-attention (MSA) and an MLP with residual connections [28] and layer normalization. The forward propagation is expressed as:

$$z'_l = \text{MSA}(\text{LN}(z_{l-1})) + z_{l-1}, \quad l = 1, \dots, L, \quad (4)$$

$$z_l = \text{MLP}(\text{LN}(z'_l)) + z'_l, \quad l = 1, \dots, L. \quad (5)$$

In Eq. (4), the input z_{l-1} is normalized and passed through MSA, after which a residual connection yields z'_l . Eq. (5) shows a second normalization, followed by the MLP and another residual addition, giving the final z_l . By stacking L layers, the ViT progressively integrates local patch information into a global representation.

2.3. Graph neural network

GNNs are designed to operate on graph-structured data, where nodes represent entities and edges capture their relations. By iteratively propagating information, GNNs learn node representations that reflect both local neighborhoods and global structure. Among the representative models, the Graph Attention Network (GAT) [29] and Graph Isomorphism Network (GIN) [30] are widely studied.

The GAT extends the Graph Convolutional Network [31] by introducing attention to weight neighbors adaptively. The embedding of node i is updated as follows:

$$h'_i = \sigma\left(\sum_{j \in N(i)} \alpha_{ij} W h_j\right), \quad (6)$$

where h_j is the feature of neighbor j , W is a learnable weight matrix, and α_{ij} is the normalized attention score. This allows the model to emphasize more informative neighbors when aggregating features.

The GIN focuses on expressive power for distinguishing graph structures. The embedding of node i is formulated as follows:

$$h'_i = \text{MLP}\left((1 + \epsilon) h_i + \sum_{j \in N(i)} h_j\right), \quad (7)$$

where ϵ is a learnable parameter controlling the balance between self-information and neighbor aggregation. The summation ensures permutation invariance, and the MLP increases expressive capacity, making GIN theoretically as powerful as the Weisfeiler–Lehman test.

2.4. Molecular fingerprint

FPs convert molecular structures into fixed-length vectors for similarity search, property prediction, and screening. By encoding chemical information in compact form, FPs are widely used in cheminformatics and drug discovery.

Functional Class Fingerprints (FCFP) [32], derived from the Morgan algorithm [33], refine atom labels by iteratively incorporating neighboring atoms and bonds. This captures both structural details and functional roles such as bond types or donor/acceptor groups.

Pharmacophore 2D (Pharm2D) [34] emphasizes pharmacological features that influence molecular activity. It represents patterns such as hydrogen bond donors/acceptors, hydrophobic regions, and charged groups in a two-dimensional framework. By focusing on these functional properties, Pharm2D provides a complementary perspective to structure-based fingerprints.

2.5. 专家混合

MoE（专家混合模型）是一种先进的架构范式，通过条件计算提高大规模模型的效率，在这种方式中，输入数据会选择性地被多个专门的专家网络处理。这种方法使模型能够高效扩展，同时减少计算资源的使用，因此对于处理高维数据和复杂关系特别有效。MoE 已在大型语言模型、自然语言处理和计算机视觉领域展示了最先进的性能。对于需要处理海量数据同时保持计算效率的任务，MoE 尤其具有优势。根据输入数据如何路由到专家网络，MoE 可以分为稠密 MoE 和稀疏 MoE。

2.5.1. 密集专家模型 (MoE)

密集专家模型（Dense MoE）会同时激活所有专家网络，将它们的输出聚合成一个统一的表示，涵盖多样化的视角和复杂特征。密集专家模型的输出公式如下：

$$F_{\text{MoE}}^{\text{dense}}(x; \Theta, \{W_i\}_{i=1}^N) = \sum_{i=1}^N G(x; \Theta)_i f_i(x; W_i) \quad (8)$$

$$G(x; \Theta)_i = \text{softmax}(g(x; \Theta))_i \quad (9)$$

其中 x 表示输入数据， Θ 表示门控网络的参数， W_i 是第 i 个专家网络的参数。项 $G(x; \Theta)_i$ 对应于门控网络计算的权重，该权重根据输入 x 动态调节每个专家的贡献。 $f_i(x; W_i)$ 是第 i 个专家对输入 x 的输出， $g(x; \Theta)_i$ 是第 i 个专家的原始门控分数。这些门控权重使用 softmax 函数进行归一化，使得 $\sum_{i=1}^N G(x; \Theta)_i = 1$ 。在训练过程中，门控函数学习为最相关的专家分配更高的权重，从而确保模型有效利用它们的专业知识。

2.5.2. 稀疏 MoE

稀疏 MoE 使用一种 top- K 路由策略，其中每个输入只激活一部分专家，从而降低计算成本。其计算定义如下：

$$G(x; \Theta)_i = \text{softmax}(\text{Top_K}(g(x; \Theta), K))_i \quad (10)$$

在这里，Top- K 函数保留 K 个最大的门控分数，并将其余的设置为一。在经过 softmax 归一化后，只有选定的 K 个专家会收到非零权重。最终输出如公式(8)所示计算，但仅限于选定的专家。这样的选择性路由大大减少了计算量，同时保持了模型容量，使稀疏 MoE 特别适合处理复杂模式。

3. 方法

在本节中，我们介绍了一个用于处理分子数据以预测药物毒性的多模态框架。图1-(a)展示了MEMOL完整工作流程的概览。模型的输入是分子的简化分子线性输入系统（SMILES）字符串 [35]，通过 RDKit [34]，从中派生出图像和图模态，而 FP 模态则使用 MolFeat [36] 生成。为了有效提取代表性特征，采用优化的特征提取技术获得模态特定的嵌入向量。然后，这些嵌入向量通过 MoE 自注意力模块处理，该模块将多头自注意力与 MoE 结合，使模型能够捕捉模态内部的依赖关系。随后，嵌入向量通过 MoE 交叉注意力模块，在其中将 MCA 与 MoE 结合以促进模态间的交互。MEMOL 首先学习图像和图模态之间的关系，然后整合它们。

3.1. 特征嵌入

3.1.1. 图像嵌入

使用 RDKit 库，从 SMILES 字符串生成了具有三个通道、大小为 224×224 的药物分子图像，并将其作为预训练 ViT 模型的输入。该架构利用了 ViT 在大规模数据集上预训练的特征表示能力，同时允许针对特定下游任务进行微调。特征提取是通过之前在公式 (4) 和 (5) 中描述的转换过程进行的。

3.1.2. 图嵌入

为了有效地处理图结构，MEMOL 使用了 GIN 和 GAT，如第 2.3 节所述。具体而言，在 GIN-GAT-GIN-GAT 的顺序中依次排列了两个 GIN 层和两个 GAT 层，从分子图中提取特征。这种混合方法利用了两种模型的优势：GIN 能够有效地学习具有鉴别性的节点级表示，而 GAT 则通过注意力机制自适应地对邻居节点的贡献进行加权。最后，为了将提取的节点级特征整合为图级表示，对所有节点嵌入进行全局平均池化。

3.1.3. 指纹嵌入

FCFP 和 Pharm2D 由 MolFeat 生成，用于 MEMOL 中的分子指纹。由于这些指纹具有高维特性，因此设计并预训练了一个两层自编码器，以有效压缩和编码其关键特征。自编码器由编码器-解码器结构组成，其中编码器将输入的指纹映射到紧凑的潜在空间，解码器则将压缩后的特征重建回原始输入。使用自编码器的优势不仅在于降维，还在于保留指纹中最具信息量的特征。预训练后，只有自编码器的编码器部分被集成到管道中。

在预训练阶段，自编码器在无监督环境下进行训练。具体来说，输入与输出之间的均方误差被用作重构损失，以确保学习到的潜在空间能够保留最突出的分子特征。自编码器的输入是一个 2048 维的向量，由 1024 维的 FCFP 和 Pharm2D 拼接而成。编码器由两层全连接层组成，逐步将输入压缩为 256 维的潜在表示。相反，解码器采用两层全连接层，将压缩后的表示恢复到原始的 2048 维空间。

3.2. MoE 注意力模块

图1-(b)和(c)分别展示了 MoE 自注意力模块和 MoE 交叉注意力模块。这两个模块都将 MoE 与多头注意力相结合，能够实现动态专家选择，同时有效捕捉上下文依赖。两者机制的主要区别在于其输入配置：

- MoE 自注意力处理单个输入 X ，使用它生成查询、键、值、门控网络以及残差连接。
- MoE 交叉注意力接受两个输入， X 和 Y ，其中查询和残差连接来自 X ，而键、值和门控网络则来自 Y 。

特别是在我们的模型中，通过交换 X 和 Y 对 MoE 交叉注意力执行两次，以双向捕捉关系。例如，如果我们计算图像和图图模态之间的 MoE 交叉注意力，我们首先将 X 和 Y 分别设置为图像和图，然后再交换它们，将 X 和 Y 分别设置为图和图像。此外，由于 X 通过残差连接保留其信息，从 Y 生成门控网络可以防止

2.5. Mixture of experts

The MoE is an advanced architectural paradigm that enhances the efficiency of large-scale models through conditional computation, where input data is selectively processed by multiple specialized expert networks. This approach allows the model to scale efficiently while reducing computational resources, making it particularly effective for handling high-dimensional data and intricate relationships. MoE has demonstrated state-of-the-art performance in the fields of large language models, natural language processing, and computer vision. It is especially advantageous for tasks that require processing vast amounts of data while maintaining computational efficiency. MoE can be categorized into dense MoE and sparse MoE based on how the input data are routed to expert networks.

2.5.1. Dense MoE

Dense MoE activates all expert networks simultaneously, aggregating their outputs into a unified representation that encapsulates diverse perspectives and complex features. The output of dense MoE is formulated as follows:

$$F_{\text{MoE}}^{\text{dense}}(x; \Theta, \{W_i\}_{i=1}^N) = \sum_{i=1}^N G(x; \Theta)_i f_i(x; W_i) \quad (8)$$

$$G(x; \Theta)_i = \text{softmax}(g(x; \Theta))_i \quad (9)$$

where x represents the input data, Θ denotes the parameters of the gating network, and W_i is the parameter of the i th expert network. The term $G(x; \Theta)_i$ corresponds to the gating network's computed weight, which dynamically modulates the contribution of each expert based on the input x . $f_i(x; W_i)$ is the output of the i th expert for input x and $g(x; \Theta)_i$ is the raw gate score of i th expert. These gating weights are normalized using the softmax function, such that $\sum_{i=1}^N G(x; \Theta)_i = 1$. During training, the gating function learns to assign higher weights to the most relevant experts, ensuring that the model efficiently leverages their specialized knowledge.

2.5.2. Sparse MoE

Sparse MoE uses a top- K routing strategy, where only a subset of experts is activated for each input, thus lowering computational cost. Its computation is defined as:

$$G(x; \Theta)_i = \text{softmax}(\text{Top_K}(g(x; \Theta), K))_i \quad (10)$$

Here, the Top- K function keeps the K largest gate scores and sets the rest to $-\infty$. After softmax normalization, only the selected K experts receive nonzero weights. The final output is then computed as in Eq. (8), but restricted to the selected experts. This selective routing greatly reduces computation while preserving model capacity, making sparse MoE particularly suitable for complex patterns.

3. Methods

In this section, we introduce a multimodal framework for processing molecular data to predict drug toxicity. Fig. 1-(a) provides an overview of the complete workflow of MEMOL. The input to the model is the Simplified Molecular Input Line Entry System (SMILES) [35] string of a molecule, from which image and graph modalities are derived using RDKit [34], while the FP modality is generated using MolFeat [36]. To extract representative features effectively, modality-specific embedding vectors are obtained using optimized feature extraction techniques. These embeddings are then processed through the MoE self-attention block, which incorporates multi-head self-attention with the MoE, enabling the model to capture intra-modality dependencies. Subsequently, the embeddings are passed through the MoE cross-attention block, where MCA is incorporated with MoE to facilitate inter-modality interactions. MEMOL first learns the relationships between image and graph modalities, and then integrates them with FP representations. After this multimodal fusion process, the model predicts the toxicity profile of the molecular structure, classifying it as either toxic or non-toxic.

3.1. Feature embedding

3.1.1. Image embedding

Using the RDKit library, images of drug molecules with three channels and a size of 224×224 were generated from SMILES strings and used as inputs for a pre-trained ViT model. This architecture leverages the feature representation capability of ViT, pre-trained on large-scale datasets, while allowing fine-tuning for specific downstream tasks. Feature extraction is performed using the transformation process previously described in Eqs. (4) and (5).

3.1.2. Graph embedding

To effectively process the graph structure, MEMOL uses GIN and GAT, as described in Section 2.3. Specifically, two GIN layers and two GAT layers are sequentially arranged in GIN-GAT-GIN-GAT order to extract features from a molecular graph. This hybrid approach leverages the strengths of both models: GIN effectively learns discriminative node-level representations, whereas GAT adaptively weights the contributions of neighboring nodes through attention. Finally, to integrate the extracted node-level features into a graph-level representation, global mean pooling is performed over all node embeddings.

3.1.3. Fingerprint embedding

FCFP and Pharm2D, generated by MolFeat, were used in MEMOL for molecular FPs. Because these FPs have high dimensionality, a two-layer autoencoder was designed and pre-trained to effectively compress and encode their essential features. The autoencoder consists of an encoder-decoder structure, where the encoder maps input FPs into a compact latent space, and the decoder reconstructs the compressed features to the original input. The advantage of using an autoencoder is not only to reduce dimensionality but also to retain the most informative features in FPs. After pre-training, only the encoder part of the autoencoder was integrated into the pipeline.

For the pre-training phase, the autoencoder was trained in an unsupervised setting. Specifically, the mean squared error between the input and output is used as the reconstruction loss to ensure that the learned latent space preserves the most salient molecular features. The input to the autoencoder was a 2048-dimensional vector obtained by concatenating the 1024-dimensional FCFP and Pharm2D. The encoder was composed of two fully connected layers, progressively reducing the input to a 256-dimensional latent representation. Conversely, the decoder employs two fully connected layers to restore the compressed representation back to its original 2048-dimensional space.

3.2. MoE attention block

Fig. 1-(b) and (c) illustrate the MoE self-attention block and MoE cross-attention block, respectively. Both modules integrate MoE with multi-head attention, enabling dynamic expert selection while effectively capturing contextual dependencies. The primary distinction between these mechanisms lies in their input configuration:

- **MoE self-attention** processes a single input X , using it to generate the query, key, value, gating networks and residual connection.
- **MoE cross-attention** takes two inputs, X and Y , where the query and residual connection are derived from X , while the key, value and gating networks come from Y .

In particular, in our model, MoE cross-attention is performed twice by swapping X and Y to capture relationships bi-directionally. For example, if we compute MoE cross-attention between image and graph modalities, we first set X and Y as the image and graph, respectively, and then we exchange them, setting X and Y as the graph and image, respectively. Furthermore, since X retains its information through the residual connection, generating the gating network from Y prevents

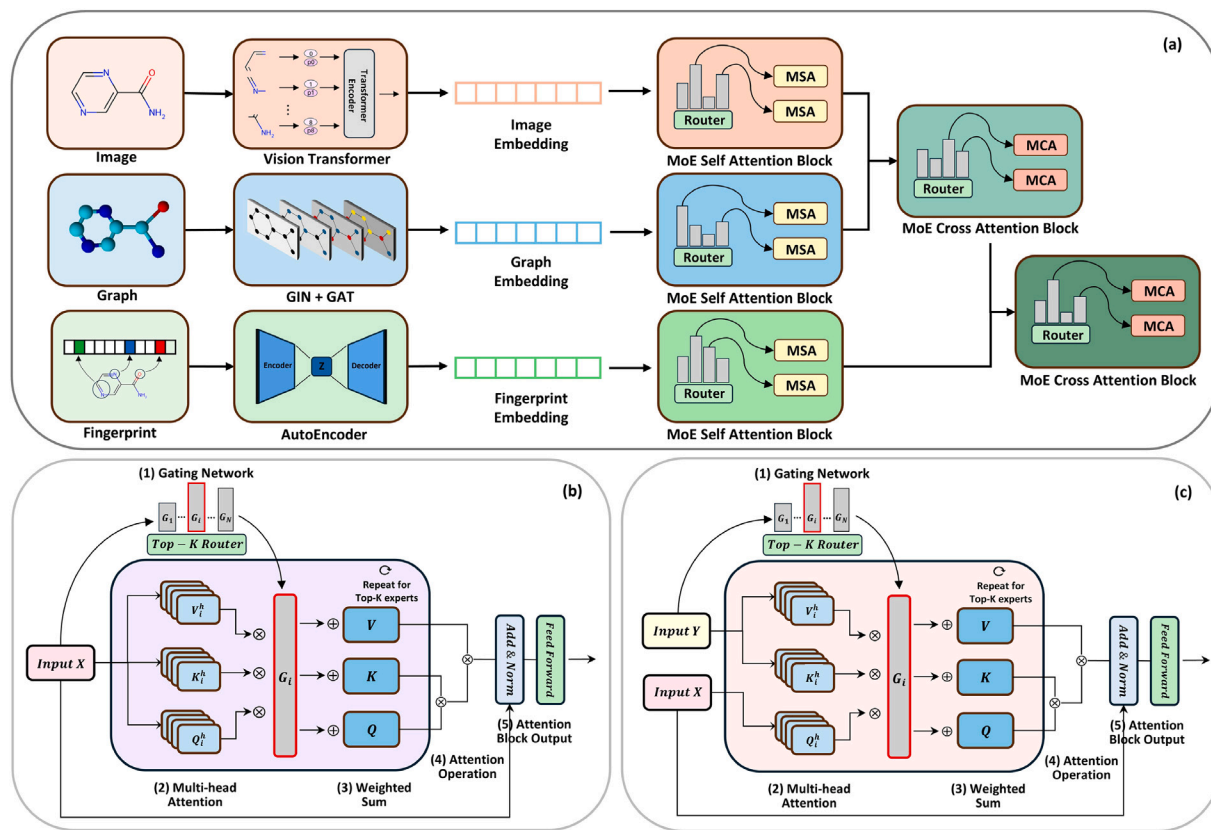


图 1. Illustration of MEMOL architecture. (a) shows the overall framework. MEMOL processes three input modalities – image, graph, and FP – using a ViT, a GIN-GAT hybrid model, and an Autoencoder, respectively. The extracted embeddings pass through MoE self-attention blocks,

followed by MoE cross-attention, and blocks to integrate modalities. (b) and (c) represent an MoE self-attention block and an MoE cross-attention block, respectively. (b) processes a single input X

(c) processes two inputs X and Y . In the MoE attention block, following steps are sequentially carried out: (1) A gating network selects top K experts based on computed scores, where the experts are generated from input Y . (2) Each selected expert independently performs multi-head attention, generating query,

key, and value representations. (3) Expert outputs are aggregated via weighted summation. (4) Attention operation is performed using the accumulated Q , K , and V representations. (5) A feed-forward network with residual connections and layer normalization enhances stability.

and V representations. (5) A feed-forward network with residual connections and layer normalization enhances stability.

过分强调 X 的原始信息。相反，它允许模型灵活调整 X 与来自 Y 的新信息的融合方式，从而实现两种输入之间更有效的整合。

尽管输入结构存在差异，但两种机制共享第 2.5.2 节中描述的相同计算过程。在自注意力情况下，一旦门控分数确定后，查询、键和值矩阵将按如下方式生成：

$$Q^{i,h} = XW_Q^{i,h}, \quad K^{i,h} = XW_K^{i,h}, \quad V^{i,h} = XW_V^{i,h} \quad (11)$$

$i \in S$ 表示所选专家的索引， $h \in \{1, \dots, H\}$ 表示多头索引，而 Q^i 、 K^i 、 V^i 、 h 分别表示第 i 个专家的第 h 个头的查询、键和值矩阵。 X 指输入数据，而 $W_Q^{i,h}$ 、 $W_K^{i,h}$ 以及 $W_V^{i,h}$ 是可学习的权重矩阵。在交叉注意力情况下，唯一的区别是 $K^{i,h} = YW_K^{i,h}$ 和 $V^{i,h} = YW_V^{i,h}$ 。

随后，门控分数 G_i 对每个专家在注意力头上的输出进行加权。最终的加权求和如下所示：

$$Q^h = \sum_{i=1}^S G_i \cdot Q^{i,h}, \quad K^h = \sum_{i=1}^S G_i \cdot K^{i,h}, \quad V^h = \sum_{i=1}^S G_i \cdot V^{i,h}, \quad (12)$$

对于选定的 S 专家，所有计算值都会被累积，并使用多头注意力机制计算注意力值 X_{attn} ，如公式 (2)、(3) 所定义。将稀疏 MoE 与注意力机制结合，使得通过动态选择最相关的专家可以实现更高效、专业的特征提取。然后，最终输出会经过残差连接和层归一化处理，

前馈网络（FFN）和 dropout，其公式如下：

$$X_{\text{out}} = \text{Dropout}(\text{FFN}(\text{LayerNorm}(X + X_{\text{attn}}))) \quad (13)$$

残差连接通过保留原始输入同时缓解梯度消失问题，从而稳定训练。层归一化对特征分布进行规范化，提高学习效率并加速收敛。前馈神经网络（FFN）由两个带 ReLU 激活函数的全连接层组成，引入非线性以增强模型的表达能力。最后，Dropout 通过在训练过程中随机停用神经元来防止过拟合，从而提高模型的泛化性能。

4. 结果

4.1 数据集与实验设置

本研究使用来自治疗数据共享平台（Therapeutic Data Commons, TDC）[37] 的数据集以及从以往研究中获得的另外两个数据集。TDC 数据库提供了 66 个可供 AI 使用的数据集，这些数据集因其可靠性而被广泛认可，可直接用于模型训练而无需预处理。这些数据集支持多种计算任务，包括药物-药物相互作用、药物-靶点相互作用、药代动力学和毒性预测，使其成为 AI 驱动药物发现的重要资源。在本研究中，我们从 TDC 中选择了三个数据集：

- hERG 评估药物对人类 Ether-à-go-go 相关基因（hERG）通道的影响，这是心脏毒性评估中的一个关键因素。

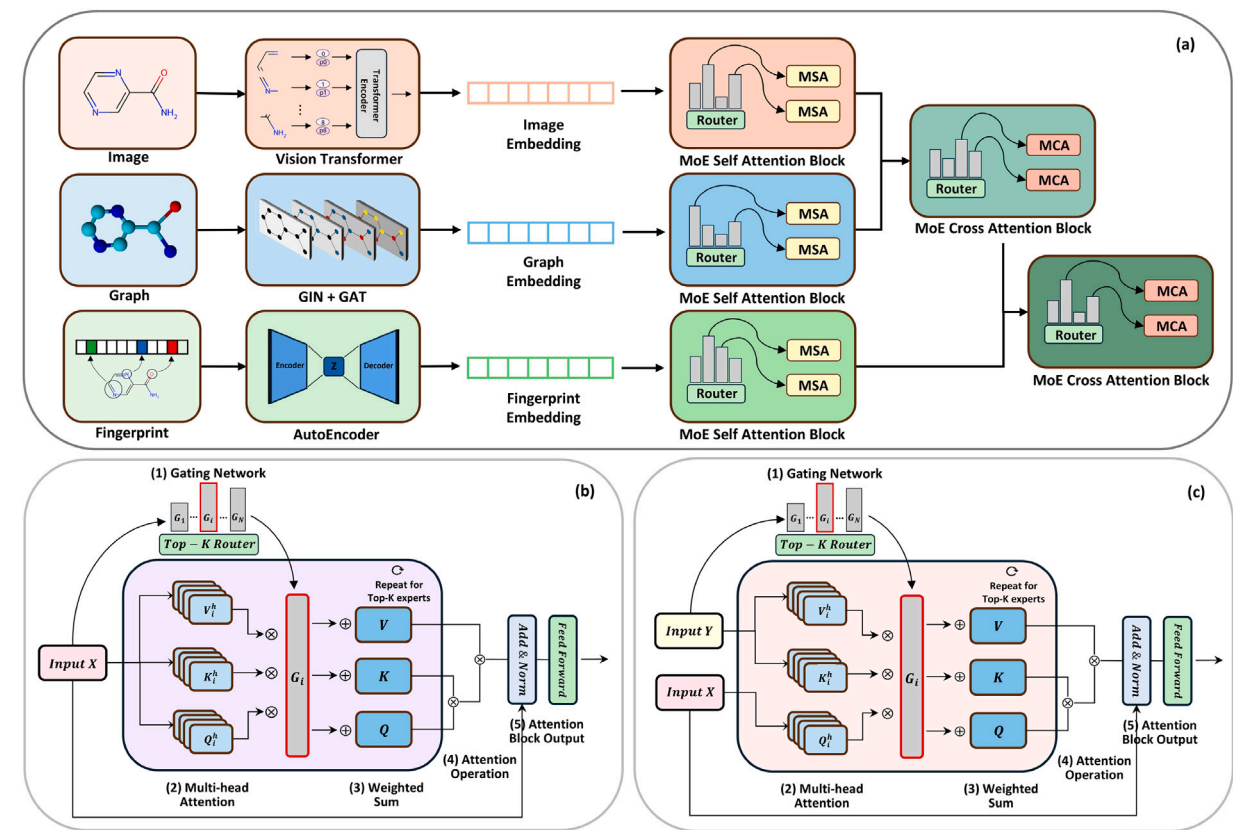


Fig. 1. Illustration of MEMOL architecture. (a) shows the overall framework. MEMOL processes three input modalities – image, graph, and FP – using a ViT, a GIN-GAT hybrid model, and an Autoencoder, respectively. The extracted embeddings pass through MoE self-attention blocks, followed by MoE cross-attention blocks to integrate modalities. (b) and (c) represent an MoE self-attention block and an MoE cross-attention block, respectively. (b) processes a single input X whereas (c) processes two inputs X and Y . In the MoE attention block, following steps are sequentially carried out: (1) A gating network selects top K experts based on computed scores, where the experts are generated from input Y . (2) Each selected expert independently performs multi-head attention, generating query, key, and value representations. (3) Expert outputs are aggregated via weighted summation. (4) Attention operation is performed using the accumulated Q , K , and V representations. (5) A feed-forward network with residual connections and layer normalization enhances stability.

excessive emphasis on the original information of X . Instead, it allows the model to flexibly adjust how X is fused with new information from Y , enabling more effective integration between the two inputs.

Despite this difference in the input structure, both mechanisms share the identical computational process described in Section 2.5.2. In the self-attention case, once the gate scores have been determined, query, key, and value matrices are generated as follows:

$$Q^{i,h} = XW_Q^{i,h}, \quad K^{i,h} = XW_K^{i,h}, \quad V^{i,h} = XW_V^{i,h} \quad (11)$$

where $i \in S$ represents the index of the selected experts, $h \in \{1, \dots, H\}$ denotes the multi-head index, and $Q^{i,h}$, $K^{i,h}$, $V^{i,h}$ represent the query, key and value matrices, respectively, from the h th head of the i th expert. X refers to the input data and $W_Q^{i,h}$, $W_K^{i,h}$, and $W_V^{i,h}$ are the learnable weight matrices. In the cross-attention case, the only difference is that $K^{i,h} = YW_K^{i,h}$ and $V^{i,h} = YW_V^{i,h}$.

Subsequently, the gate score G_i weights the outputs of each expert across attention heads. The final weighted summation is given by:

$$Q^h = \sum_{i=1}^S G_i \cdot Q^{i,h}, \quad K^h = \sum_{i=1}^S G_i \cdot K^{i,h}, \quad V^h = \sum_{i=1}^S G_i \cdot V^{i,h}, \quad (12)$$

For the selected S experts, all computed values are accumulated, and the attention value X_{attn} is computed using the multi-head attention mechanism, as defined in Eqs. (2), (3). This integration of sparse MoE with attention enables more efficient and specialized feature extraction by dynamically selecting the most relevant experts. The final output then undergoes a residual connection, layer normalization,

Feed-Forward Network (FFN), and dropout, as formulated as follows:

$$X_{\text{out}} = \text{Dropout}(\text{FFN}(\text{LayerNorm}(X + X_{\text{attn}}))) \quad (13)$$

The residual connection stabilizes training by preserving the original input while mitigating the vanishing gradient problem. Layer normalization normalizes the feature distributions, improving learning efficiency and accelerating convergence. The FFN, which is composed of two fully connected layers with a ReLU activation function, introduces non-linearity to enhance the expressive power of the model. Finally, dropout prevents overfitting by randomly deactivating neurons during training, thereby improving the generalization performance.

4. Results

4.1. Datasets and experimental settings

This study employs datasets from the Therapeutic Data Commons (TDC) [37] and two additional datasets obtained from previous research. The TDC repository provides 66 AI-ready datasets that are widely recognized for their reliability, enabling direct application to model training without preprocessing. These datasets facilitate various computational tasks, including drug-drug interactions, drug-target interactions, pharmacokinetics, and toxicity prediction, making them fundamental resources in AI-driven drug discovery. For this study, we selected three datasets from TDC:

- hERG evaluates the effects of a drug on the human Ether-à-go-go-Related Gene (hERG) channel, a critical factor in cardiac toxicity assessment.

number of compounds in each dataset before and after preprocessing , , , along with their distribution across training validation and testing sets.					
数据集 name	原始 count	处理后 count	训练 验证	Test	
hERG	655	643	449	64	130
DILI	475	457	321	47	89
皮肤 Reaction	404	404	283	40	81
Carci 某人 ici	863	768	616	76	76
谢露 DILlst	1279	1097	874	112	111

• DILI 判断药物是否会引起药物性肝损伤（DILI），这是毒性预测中的一个主要挑战。• 皮肤反应记录不良皮肤效应，包括刺激和过敏反应。

此外，我们在实验中还整合了文献中的两个数据集，列在下面：

• 致癌性 [38] 包含被归类为致癌或非致癌的化学化合物，这些数据来源于国家毒理学研究中心的肝癌数据库。• DILlst [39] 是FDA DILlrnk 数据集的扩展版本，整合了来自文献的额外数据，以改善DILI的预测。

在预处理过程中，通过去除盐类来优化 SMILES 字符串，这些盐类在记号中使用句点（ ‘.’ ）分隔符被识别为独立片段。盐类是通常为提高溶解度或稳定性而添加的对离子，但保留它们可能会引入冗余的分子表示，导致潜在的数据噪声。此外，FP 转换过大、图表示过小或无机化合物也被排除。经过这些预处理步骤后，最终数据集的结构如表 1 所概述。由于 TDC 库提供了预先拆分的数据集，以便与其他模型进行公平比较，我们遵循了原始的划分策略，并按 7:1:2 的比例进行训练、验证和测试。相比之下，直接从文献中获取的致癌性和 DILlst 数据集则使用 8:1:1 的比例进行划分。在模型训练中，我们使用了 Adam 优化器，并进行了广泛的超参数调优以进行

为了评估模型的性能，本研究采用受试者工作特征曲线下面积（AUROC）和精确率-召回率曲线下面积（AUPRC）作为评估指标。AUROC 通过分析在不同阈值下的真正率（TPR）与假正率（FPR）之间的权衡来衡量分类性能。TPR 表示在所有实际为正的样本中被正确识别为正的比例，而 FPR 表示被错误分类为正的负样本的比例。类似地，AUPRC 通过分析精确率与召回率之间的平衡来评估模型性能，在处理不平衡数据集时尤其有用。AUROC 和 AUPRC 都被广泛应用于分类任务中，为模型的稳健性和可靠性提供全面的评估。

4.2. 比较研究

为了验证MEMOL的有效性，我们将其性能与五种传统机器学习模型和七种深度学习模型进行了比较。传统的机器学习模型包括随机森林（RF）、极端梯度提升（XGBoost）、轻量级梯度提升机（LGBM）、支持向量机（SVM）和线性回归（LR）。用于比较的深度学习模型如下：

• ChemBERTa [40] 是一种基于 Transformer 的模型，通过在大规模 SMILES 字符串上进行预训练来进行分子性质预测。利用 RoBERTa [46] 架构，它通过掩码语言建模学习上下文化学表示，支持在生化和药理学任务上的强迁移能力。

• MolCLR [41] 是一个自监督学习框架，使用图神经网络（GNN）通过对比学习在大规模分子数据上进行预训练。它生成正负分子对，从而提升表示学习能力和下游分子性质预测的性能。

• GraphMVP [42] 是一种多视图预训练模型，结合了二维分子图和三维分子几何结构。它通过整合拓扑信息和空间信息来学习更丰富的分子表征，即使在缺乏明确三维数据的任务中也能发挥有效作用。

• MolFormer [43] 是一种基于Transformer的模型，使用超过11亿个未标注分子的SMILES序列进行预训练。它采用线性注意力和旋转位置嵌入来增强分子结构学习，特别用于结构-性质关系的预测。

• GIT-Mol [13] 是一个多模态大型语言模型，整合了分子科学中的图、图像和文本模态。在数百万分子上进行预训练，它在分子性质预测、图像描述以及分子生成任务中取得了显著的提升。

• AMPred-CNN [44] 是一个基于卷积神经网络的模型，用于处理二维分子图像。它通过卷积层提取结构模式，利用基于图像的特征表示进行毒性预测。• CToxPred2 [45] 是一个心脏毒性预测模型，在半监督学习框架下使用分子指纹和描述符。该模型在200万未标记分子上进行训练，采用伪标签和子模优化以提高预测准确性。

表2比较了各种模型在之前描述的五个百分点上的表现，基于AUROC和AUPRC指标。相比于在hERG数据集上表现第二的模型CToxPred2，MEMOL的AUROC提高了3.6%，AUPRC提高了0.44%。在DILI数据集上，MEMOL的AUROC比MolFormer高出2.64%，AUPRC高出1.89%。在皮肤反应数据集上，MEMOL取得了最高的AUROC，比GIT-Mol提高了2.13%，同时AUPRC也比GraphMVP高1.87%。同样，在致癌性数据集上，MEMOL取得了最高的AUROC，比MolFormer高出3.02%。在DILlst数据集上，MEMOL的AUROC比GIT-Mol提高了8.33%，AUPRC比GraphMVP提高了9.11%。

虽然在致癌性任务上，MEMOL 的 AUPRC 较其他三种模型略低，但它仍显示出具有竞争力的性能。总体而言，MEMOL 在除一个任务之外的所有情况下都取得了最佳表现，并且比第二佳模型的性能提升了最高 9.11%，突显了其在多样化任务中的卓越预测能力。此外，我们还对所有数据集进行了统一划分比例的进一步实验，具体细节见补充材料表 S2。我们还进行了统计显著性分析，以验证 MEMOL 相对于基线模型改进的稳健性。详细的方法和结果见补充材料（参见图 S1-S5）。

4.3. MoE 中参数分析

为了评估稀疏 MoE 中专家总数（ E ）以及通过 top K 选择的专家数量（ K ）的影响，我们进行了实验，通过改变 E 的值，同时将 K 设置为 1 和 2，这些是 top K 选择中常用的选项 [47]。具体而言， E 被配置为 {2, 4,

The number of compounds in each dataset before and after preprocessing, along with their distribution across training, validation, and testing sets.					
Dataset name	Raw count	Processed count	Train	Valid	Test
hERG	655	643	449	64	130
DILI	475	457	321	47	89
Skin Reaction	404	404	283	40	81
Carcinogenicity	863	768	616	76	76
DILlst	1279	1097	874	112	111

• DILI determines whether a drug induces Drug-Induced Liver Injury (DILI), a major challenge in toxicity prediction. • Skin Reaction captures adverse dermatological effects, including irritation and allergic responses.

Additionally, two datasets from the literature were incorporated into our experiments, listed below:

• Carcinogenicity [38] contains chemical compounds classified as carcinogenic or non-carcinogenic, sourced from the National Center for Toxicological Research Liver Cancer Database. • DILlst [39] is an extended version of the FDA DILlrnk dataset that incorporates additional data from the literature to improve the prediction of DILI.

During preprocessing, SMILES strings were refined by removing salts, which were identified as separate fragments in the notation using a period (‘.’) delimiter. Salts are counter-ions commonly added to enhance solubility or stability, but retaining them could introduce redundant molecular representations, leading to potential data noise. Additionally, compounds too large for FP conversion, too small for graph representation, or inorganic were excluded. Following these preprocessing steps, the final dataset was structured as summarized in Table 1. Since the TDC library provides pre-split datasets to facilitate fair comparison with other models, we followed the original partitioning strategy and used a 7:1:2 ratio for training, validation, and testing. In contrast, the Carcinogenicity and DILlst datasets, which were obtained directly from the literature, were divided using an 8:1:1 ratio. For model training, we used the Adam optimizer and conducted extensive hyperparameter tuning to enhance performance. The details of the explored ranges and the final configurations for each dataset are provided in Table S1 of the Supplementary Materials.

To assess the performance of the model, this study employs the area under the receiver operating characteristic curve (AUROC) and the area under the precision–recall curve (AUPRC) as evaluation metrics. AUROC measures classification performance by analyzing the trade-off between the true positive rate (TPR) and the false positive rate (FPR) across various thresholds. TPR represents the proportion of correctly identified positive samples among all actual positives, whereas FPR denotes the proportion of negative samples misclassified as positives. Similarly, AUPRC evaluates model performance by analyzing the balance between precision and recall, making it particularly useful for imbalanced datasets. Both AUROC and AUPRC are widely adopted in classification tasks, providing a comprehensive assessment of model robustness and reliability.

4.2. Comparative study

To validate the effectiveness of MEMOL, we compared its performance with five conventional machine learning models and seven deep learning models. The conventional machine learning models include Random Forest (RF), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LGBM), Support Vector Machine (SVM), and Linear Regression (LR). The deep learning models considered for comparison are as follows:

• ChemBERTa [40] is a transformer-based model designed for molecular property prediction by pre-training on large-scale SMILES string. Leveraging RoBERTa [46] architecture, it learns contextualized chemical representations through masked language modeling, supporting strong transfer to biochemical and pharmacological tasks. • MolCLR [41] is a self-supervised learning framework that uses GNNs to pre-train on large-scale molecular data via contrastive learning. It generates positive and negative molecular pairs, improving representation learning and downstream molecular property prediction. • GraphMVP [42] is a multi-view pre-training model that combines 2D molecular graphs and 3D molecular geometric structures. It learns richer molecular representations by integrating topological and spatial information, making it effective even in tasks lacking explicit 3D data. • MolFormer [43] is a transformer-based model pre-trained on over 1.1 billion unlabeled molecules using SMILES sequences. It applies linear attention and rotary positional embeddings to enhance molecular structure learning, particularly for structure–property relationship prediction. • GIT-Mol [13] is a multimodal large language model that integrates graph, image, and text modalities for molecular science. Pre-trained on millions of molecules, it achieves notable improvements in molecular property prediction, captioning, and molecule generation tasks. • AMPred-CNN [44] is a model based on convolutional neural network to process 2D molecular images. It extracts structural patterns using convolutional layers, leveraging image-based feature representations for toxicity prediction. • CToxPred2 [45] is a cardiotoxicity prediction model that utilizes molecular FPs and descriptors within a semi-supervised learning framework. Trained on 2 million unlabeled molecules, it employs pseudolabeling and submodular optimization to enhance predictive accuracy.

Table 2 compares the performance of various models across five datasets previously described based on AUROC and AUPRC. MEMOL achieved 3.6% higher AUROC and 0.44% higher AUPRC compared to CToxPred2, the second-best performing model on the hERG dataset. For the DILI dataset, MEMOL outperformed MolFormer by 2.64% in AUROC and 1.89% in AUPRC. For the Skin Reaction dataset, MEMOL achieved the highest AUROC, showing a 2.13% improvement over GIT-Mol, and also recorded a 1.87% higher AUPRC compared to GraphMVP. Similarly, for the Carcinogenicity dataset, MEMOL recorded the highest AUROC, outperforming MolFormer by 3.02%. For the DILlst dataset, MEMOL demonstrated an 8.33% improvement in AUROC compared to GIT-Mol and a 9.11% increase in AUPRC over GraphMVP.

Although MEMOL showed lower AUPRC compared with the three other models on the Carcinogenicity, it still indicated a competitive performance. Overall, MEMOL achieved the best performance in all cases except one and outperformed the second-best model by up to 9.11%, highlighting its remarkable predictive capability across diverse tasks. In addition, we conducted further experiments with unified split ratios across all datasets, the details of which are provided in Table S2 of the Supplementary Materials. We also performed statistical significance analysis to verify the robustness of MEMOL’s improvements over baseline models. The detailed methodology and results are provided in the Supplementary Materials (see Figure S1-S5).

4.3. Analysis of parameters in MoE

To evaluate the impact of the total number of experts (E) and selected experts by top K (K) in sparse MoE, we conducted experiments by varying E while setting K to 1 and 2, which are commonly used choices for top K selection [47]. Specifically, E was configured as {2, 4,

表 2 性能

comparison of 13 predictive models across five toxicity benchmark datasets. The models were evaluated using AUROC and AUPRC metrics. The first , five models listed are conventional machine learning methods while the next seven are deep learning-based approaches.

D数据集	hERG		DILI		Skin 反应 ion		致癌性		DIList	
公制	受试者工作特征曲线 线下面积	AUPRC	受试者工作特征曲线 线下面积	AUPRC	受试者工作特征曲线 线下面积	AUPRC	受试者工作特征曲线 线下面积	AUPRC	受试者工作特征曲线 线下面积	AUPRC
RF	0.8455	0.9349	0.8721	0.8586	0.6706	0.7437	0.7120	0.7762	0.7271	0.6546
LGBM	0.8270	0.9330	0.8812	0.8975	0.6821	0.7303	0.5833	0.6837	0.6787	0.6892
XGB	0.8347	0.9408	0.8878	0.9085	0.7073	0.7729	0.5355	0.6248	0.7300	0.7439
SVM	0.7132	0.8435	0.3868	0.4896	0.7806	0.8016	0.6848	0.7631	0.6683	0.6766
LR	0.4243	0.7289	0.8094	0.7962	0.5957	0.7025	0.3826	0.5392	0.461	0.5085
化工与生物工程 RTa [40]	0.8272	0.9202	0.8873	0.9051	0.7749	0.8336	0.6986	0.8307	0.6949	0.6992
Mol清除 [41]	0.8428	0.9364	0.9080	0.9152	0.7264	0.7905	0.6304	0.7726	0.6862	0.7006
GraphMVP [42]	0.7449	0.8752	0.8954	0.9237	0.7812	0.8808	0.7087	0.8288	0.7128	0.7488
MolFormer [43]	0.8356	0.9369	0.9211	0.9342	0.8042	0.8596	0.7217	0.8411	0.7138	0.7228
GIT-Mol [13]	0.8441	0.9386	0.9060	0.9085	0.8093	0.8774	0.7116	0.8047	0.7303	0.7264
AMPred-CNN [44]	0.7959	0.9152	0.7750	0.7839	0.7079	0.7868	0.6681	0.7550	0.6598	0.6346
CToxPred2 [45]	0.8691	0.9530	0.8721	0.9025	0.7526	0.8061	0.6924	0.7429	0.7149	0.6870
备忘录	0.9004	0.9572	0.9454	0.9519	0.8265	0.8973	0.7435	0.8140	0.7911	0.8170

表格 3 AUROC results of MEMOL with varying numbers of experts and top K expert selection across five toxicity , benchmark datasets. The column K represents the number of experts selected via top K routing

with K set , to either 1 or 2. The column E denotes the total number of experts in the model

which varies among {2, 4, 8, , , 16, 32}. The proposed configuration where $K = 2$ and $E = 4$

all expert configurations. achieved the most balanced performance across						
TopK (K)	编号 of experts (E)	hERG	DILI	皮肤 r eaction	致癌性	DIList
1	2	0.8863	0.9161	0.7710	0.7145	0.7118
	4	0.8997	0.9434	0.8361	0.7507	0.7537
	8	0.8752	0.9024	0.7819	0.7087	0.7368
	16	0.8835	0.9009	0.7742	0.7391	0.7385
	32	0.8704	0.8933	0.7851	0.6884	0.7333
2	2	0.8787	0.9272	0.7647	0.7014	0.7557
	4	0.9004	0.9454	0.8265	0.7435	0.7911
	8	0.8838	0.9075	0.7940	0.7319	0.7560
	16	0.8831	0.9277	0.7883	0.6819	0.7235
	32	0.8796	0.9232	0.8112	0.7094	0.7560

8、16、32，以及前面实验中使用的相同五个毒性基准数据集用于评估。每种配置的性能使用AUROC指标进行衡量，如表3所总结，并在图2中进行了可视化。

当 $K = 1$ 时，模型在 $E = 4$ 展现了所有数据集中最高的 AUROC 值。总体而言，增加 E 会导致大多数数据集的 AUROC 下降，所有情况中最低值出现在 $E = 32$ ，但皮肤反应和 DIList 数据集除外。这一趋势在致癌性数据集中尤为明显，AUROC 从 $E = 4$ 时的 0.7507 降至 $E = 32$ 时的 0.6884。

对于 $K = 2$ ，在大多数实验设置中，其整体表现都优于 $K = 1$ 。由于实验使用了五个毒性基准数据集和五种不同的专家配置，总共评估了 25 种情况。在这些情况下， $K = 2$ 在 25 种情况中的 17 种中表现优于 $K = 1$ ，展示了其在性能上的优势。这表明，在推理阶段选择两位专家可以增强模型的稳健性。

特别是，当 K 和 E 分别为 2 和 4 时，该模型在 3 个数据集上达到了最佳性能，在 2 个数据集上达到了次优性能。值得注意的是，对于 DIList 数据集，在此设置下我们的模型表现出显著提升，AUROC 达到 0.7911，比次优配置 $K = 2, E = 8$ 和 $K = 2, E = 32$ 高 4.64%。类似于在 $K = 1$ 中的观察，增加 E 并不一定会提升 $K = 2$ 的性能。虽然 $K = 2$ 通常比 $K = 1$ 产生更好的结果，但随着 E 的增加，性能仍然有所下降，这表明更多的专家并不总是有利，甚至可能引入噪声或冗余，从而限制模型的稳健性。

如图 2 所示，当 E 超过四时，所有设置的 AUROC 值都较低，而且整体趋势显示下降。

随着专家数量的增加，性能有所提升。这一趋势在 hERG 和致癌性等数据集中尤为明显，过多的专家数量会导致显著的性能下降。总之，实验表明，选择 $K = 2$ 和 $E = 4$ 可以在专家选择策略与预测性能之间提供最有效的平衡。

4.4. 多模态学习的影响

4.4.1. 模态组合的评估

我们通过对不同模态组合进行实验来评估每种模态的贡献。具体来说，我们评估了三种情景：仅使用一种模态、整合两种模态，以及融合图像、图结构和指纹（FP）三种模态。需要注意的是，当仅使用单一模态时，模型不会应用交叉注意力模块。性能使用AUROC进行测量，结果如图3所示。

当仅使用图像模态时，模型在大多数数据集上的准确率相对较低，这表明单靠分子图像无法提供足够的结构信息。然而，当从使用所有三种模态的情况下去除图像特征时，所有数据集的性能均有所下降。这表明，尽管单独的图像信息不是很有价值，但将其与其他模态结合可以提高预测的准确性。值得注意的是，当未包含图像特征时，Skin Reaction 和 DIList 数据集的准确率显著下降，显示了图像信息的互补作用。

与图像类似，当单独使用时，图表模态的表现也有限。然而，当将图表添加到集成图像和FP中时，除了hERG数据集外，所有数据集的性能都有所提高。最显著的提升出现在皮肤反应（Skin Reaction）上。

Table 2

Performance comparison of 13 predictive models across five toxicity benchmark datasets. The models were evaluated using AUROC and AUPRC metrics. The first five models listed are conventional machine learning methods, while the next seven are deep learning-based approaches.

Dataset	hERG		DILI		Skin Reaction		Carcinogenicity		DIList	
Metric	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
RF	0.8455	0.9349	0.8721	0.8586	0.6706	0.7437	0.7120	0.7762	0.7271	0.6546
LGBM	0.8270	0.9330	0.8812	0.8975	0.6821	0.7303	0.5833	0.6837	0.6787	0.6892
XGB	0.8347	0.9408	0.8878	0.9085	0.7073	0.7729	0.5355	0.6248	0.7300	0.7439
SVM	0.7132	0.8435	0.3868	0.4896	0.7806	0.8016	0.6848	0.7631	0.6683	0.6766
LR	0.4243	0.7289	0.8094	0.7962	0.5957	0.7025	0.3826	0.5392	0.461	0.5085
ChemBERTa [40]	0.8272	0.9202	0.8873	0.9051	0.7749	0.8336	0.6986	0.8307	0.6949	0.6992
MolCLR [41]	0.8428	0.9364	0.9080	0.9152	0.7264	0.7905	0.6304	0.7726	0.6862	0.7006
GraphMVP [42]	0.7449	0.8752	0.8954	0.9237	0.7812	0.8808	0.7087	0.8288	0.7128	0.7488
MolFormer [43]	0.8356	0.9369	0.9211	0.9342	0.8042	0.8596	0.7217	0.8411	0.7138	0.7228
GIT-Mol [13]	0.8441	0.9386	0.9060	0.9085	0.8093	0.8774	0.7116	0.8047	0.7303	0.7264
AMPred-CNN [44]	0.7959	0.9152	0.7750	0.7839	0.7079	0.7868	0.6681	0.7550	0.6598	0.6346
CToxPred2 [45]	0.8691	0.9530	0.8721	0.9025	0.7526	0.8061	0.6924	0.7429	0.7149	0.6870
MEMOL	0.9004	0.9572	0.9454	0.9519	0.8265	0.8973	0.7435	0.8140	0.7911	0.8170

Table 3

AUROC results of MEMOL with varying numbers of experts and top K expert selection across five toxicity benchmark datasets. The column K represents the number of experts selected via top K routing, with K set to either 1 or 2. The column E denotes the total number of experts in the model, which varies among {2, 4, 8, 16, 32}. The proposed configuration, where $K = 2$ and $E = 4$, achieved the most balanced performance across all expert configurations.

TopK (K)	Num of experts (E)	hERG	DILI	Skin reaction	Carcinogenicity	DIList
1	2	0.8863	0.9161	0.7710	0.7145	0.7118
	4	0.8997	0.9434	0.8361	0.7507	0.7537
	8	0.8752	0.9024	0.7819	0.7087	0.7368
	16	0.8835	0.9009	0.7742	0.7391	0.7385
	32	0.8704	0.8933	0.7851	0.6884	0.7333
2	2	0.8787	0.9272	0.7647	0.7014	0.7557
	4	0.9004	0.9454	0.8265	0.7435	0.7911
	8	0.8838	0.9075	0.7940	0.7319	0.7560
	16	0.8831	0.9277	0.7883	0.6819	0.7235
	32	0.8796	0.9232	0.8112	0.7094	0.7560

8, 16, 32}, and the same five toxicity benchmark datasets from the previous experiments were used for evaluation. The performance of each configuration was measured using the AUROC metric, as summarized in Table 3 and visualized in Fig. 2.

When $K = 1$, the model exhibited the highest AUROC values at $E = 4$ across all datasets. In general, increasing E resulted in a decline in AUROC for most datasets, with the lowest values occurring at $E = 32$ in all cases except for the Skin Reaction and DIList datasets. This trend was particularly evident in the Carcinogenicity dataset, where AUROC dropped from 0.7507 at $E = 4$ to 0.6884 at $E = 32$.

For $K = 2$, the overall performance was higher than that of $K = 1$ across most experimental settings. Since the experiments were conducted using five toxicity benchmark datasets and five different expert configurations, a total of 25 cases were evaluated. Among these, $K = 2$ outperformed $K = 1$ in 17 out of 25 cases, demonstrating its superiority in performance. This suggests that selecting two experts during inference enhances the robustness of the model.

Especially, when K and E are 2 and 4, respectively, the model achieved the best performance for 3 datasets, and the second-best for 2 datasets. Notably, for the DIList dataset, our model exhibited a substantial improvement in this setting, achieving an AUROC of 0.7911, which is 4.64% higher than the second-best configuration at $K = 2, E = 8$ and $K = 2, E = 32$. Similar to observation in $K = 1$, increasing E does not necessarily lead to improved performance in $K = 2$. While $K = 2$ generally led to better results than $K = 1$, performance degradation was still observed as E increased, indicating that a larger number of experts does not always contribute positively and may even introduce noise or redundancy, limiting the robustness of the model.

As illustrated in Fig. 2, AUROC values were lower for all settings where E exceeded four, and the overall trend showed a decline

in performance as the number of experts increased. This trend was particularly evident for datasets such as hERG and Carcinogenicity, where excessive expert count led to noticeable performance drops. In summary, the experiments demonstrate that selecting $K = 2$ and $E = 4$ provides the most effective balance between expert selection strategy and predictive performance.

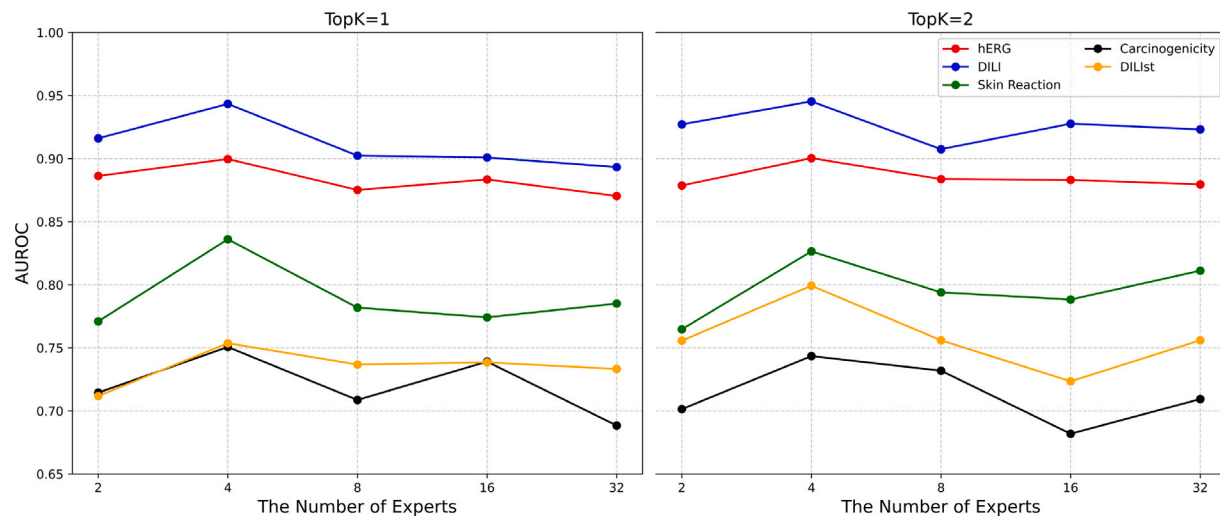
4.4. Impact of multimodal learning

4.4.1. Evaluation of modality combinations

We evaluated the contribution of each modality by conducting experiments with different combinations of modalities. Specifically, we evaluated three scenarios: using only one modality, integrating two modalities, and fusing all three modalities—image, graph, and FP. It should be noted that the model does not apply the cross-attention block when using a single modality. The performance was measured using AUROC and the results are shown in Fig. 3.

When the image modality was used alone, the accuracy of the model exhibited relatively low across most datasets, indicating that molecular images alone do not provide sufficient structural information. However, when image features were removed from the case where all three modalities were used, performance declined in all datasets. This suggests that while images alone are not highly informative, their integration with other modalities improves the accuracy of prediction. Notably, Skin Reaction and DIList datasets showed a considerable decrease when image features were not included, demonstrating their complementary role.

Similar to images, the graph modality exhibited limited performance when used alone. However, when graphs were added to the integrated image and FP, performance improved across all datasets except hERG. The most notable gains were observed in Skin Reaction,



图

2. Visualization of AUROC results of MEMOL with varying numbers of experts across five toxicity benchmark datasets. The left plot corresponds to $K = 1$, where a single expert is selected while the right plot corresponds to $K = 2$ where two experts are selected.

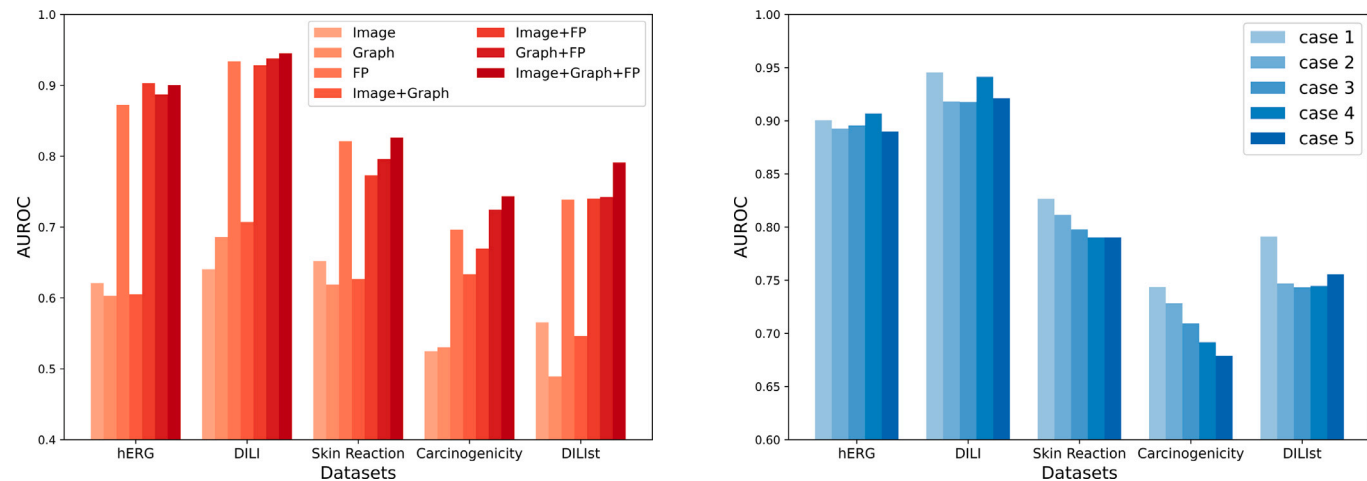


图 3. Comparison of AUROC scores for seven different modality combinations across five benchmark datasets.

致癌性，以及 DIList，表明引入基于图的表示增强了模型捕捉与毒性预测相关的结构关系和分子相互作用的能力。

即使单独使用，分子指纹（FP）的表现也很出色，说明其在捕捉相关分子特征方面具有很强的能力。然而，将分子指纹与图像和图谱相结合进一步提升了性能。当分子指纹与其他两种模态结合时，结果最为理想，hERG 除外，这突显了它们的强大但互为补充的作用。虽然单独使用分子指纹也能提供有意义的毒性预测，但通过结合图谱和图像的额外结构信息，模型的建模能力更加全面。交叉注意力权重的详细信息可参见补充材料中的图 S6。总之，整合多种分子模态使模型能够更有效地捕捉多样的结构和化学性质。

4.4.2. 模态整合顺序的评估

我们通过使用 MoE 跨注意力比较多模态融合的不同配置，分析了模态整合顺序的影响。在此，我们将 $\otimes$ 定义为 MoE 跨注意力。在本节中，我们研究了五种不同的情况，每种情况都代表一个独特的

图

4. Comparison of AUROC scores for five modality integration ordering cases across five benchmark datasets.

融合图像、图表和FP模态的方法。案例列如下：

- 案例 1 ((图像 $\otimes$ 图表) $\otimes$ FP)：图像和图表模态首先使用 MoE 跨注意力进行整合，然后是 FP，这是 MEMOL 中采用的方法。
- 案例 2 ((图像 $\otimes$ FP) $\otimes$ 图表)：图像和 FP 首先通过 MoE 跨注意力融合，然后是图表。
- 案例 3 ((图表 $\otimes$ FP) $\otimes$ 图像)：图表和 FP 首先使用 MoE 跨注意力整合，然后是图像。
- 案例 4 (拼接(图像 $\otimes$ 图表, 图像 $\otimes$ FP, 图表 $\otimes$ FP))：所有可能的模态对分别使用 MoE 跨注意力处理，然后将得到的嵌入进行拼接。这种方法消除了整合顺序的影响。
- 案例 5 (拼接(图像、图表、FP))：未应用 MoE 跨注意力；在 MoE 自注意力模块之后，简单地将三种模态拼接在一起。

图4展示了通过改变模态整合顺序在五个数据集上的实验结果。除 hERG 外，案例1始终表现出最高性能，而在 hERG 中，案例4的表现略好一些。这表明整合图像和

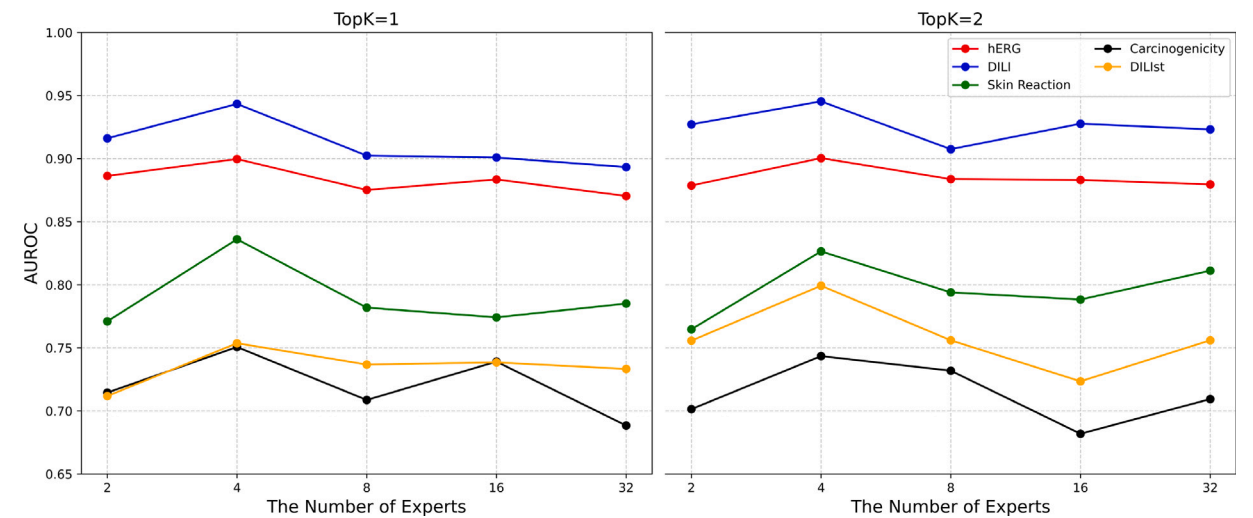


Fig. 2. Visualization of AUROC results of MEMOL with varying numbers of experts across five toxicity benchmark datasets. The left plot corresponds to $K = 1$, where a single expert is selected, while the right plot corresponds to $K = 2$, where two experts are selected.

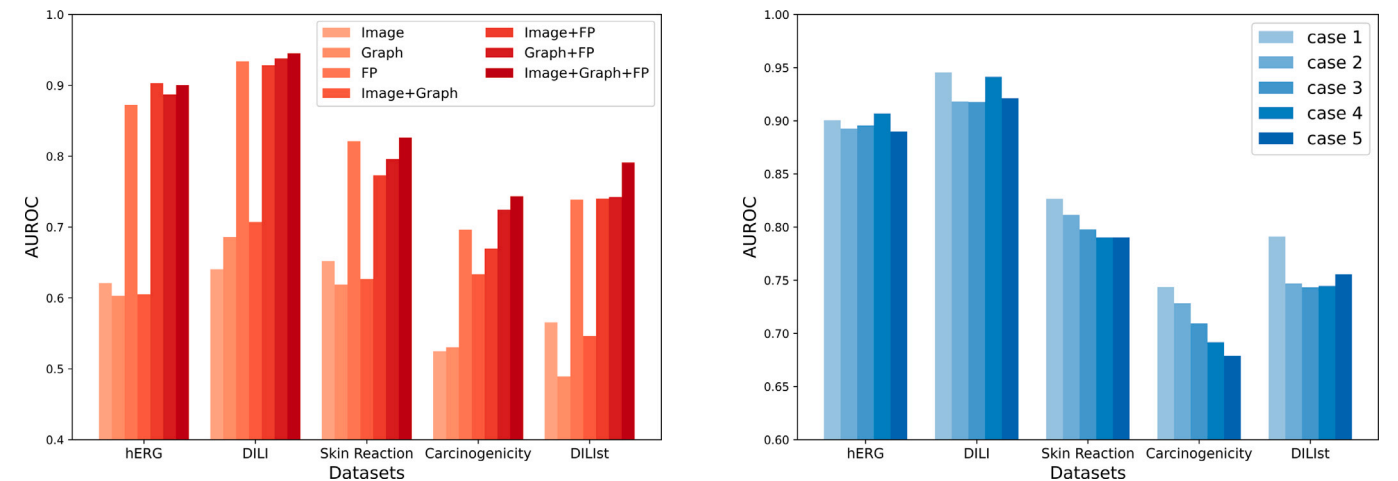


Fig. 3. Comparison of AUROC scores for seven different modality combinations across five benchmark datasets.

Carcinogenicity, and DIList, indicating that incorporating graph-based representations enhances the ability of model to capture structural relationships and molecular interactions relevant to toxicity prediction.

FP performed well even when used alone, indicating their robustness in capturing relevant molecular features. However, integrating FP with images and graphs further improved performance. The best results were consistently observed when FP was integrated with the other two modalities, except hERG, highlighting their strong but complementary role. Although utilizing FP solely provided meaningful toxicity predictions, the use of additional structural information from graphs and images resulted in more comprehensive modeling. Details of the cross-attention weights can be found in Figure S6 of the Supplementary Materials. In conclusion, integrating multiple molecular modalities enables the model to capture diverse structural and chemical properties more effectively.

4.4.2. Assessment of modality integration order

We analyzed the impact of modality integration order by comparing different configurations of multimodal fusion using MoE cross-attention. Hereafter, we define $\otimes$ as MoE cross-attention. In this section, we examined five different cases, each representing a unique

approach to fusing image, graph, and FP modalities. The cases are listed below:

- Case 1 ((Image $\otimes$ Graph) $\otimes$ FP): Image and graph modalities were first integrated using MoE cross-attention, followed by FP, which is the approach adopted in the MEMOL.
- Case 2 ((Image $\otimes$ FP) $\otimes$ Graph): Image and FP were first fused using MoE cross-attention, followed by graph.
- Case 3 ((Graph $\otimes$ FP) $\otimes$ Image): Graph and FP were first integrated using MoE cross-attention, followed by image.
- Case 4 (Concat(Image $\otimes$ Graph, Image $\otimes$ FP, Graph $\otimes$ FP)): All possible pairs of modalities were processed separately using MoE cross-attention, and the resulting embeddings were concatenated. This approach eliminates the effect of integration order.
- Case 5 (Concat(Image, Graph, FP)): No MoE cross-attention was applied; all three modalities were simply concatenated after the MoE self-attention block.

Fig. 4 presents the experimental results across five datasets by changing the order of modality integration. Case 1 consistently achieved the highest performance, except for hERG, where case 4 performed slightly better. This suggests that integrating image and

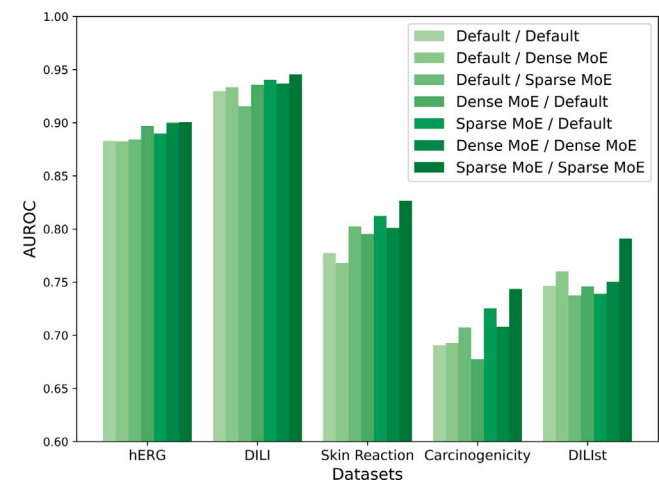


图 5. AUROC results from the ablation study of the proposed model on five benchmark datasets. In the legend, the left part represents the method applied to self-attention while the right part denotes the approach used for cross-attention.

在将 FP（在先前实验中被认定为最有信息量的模态）整合之前先进行图处理，可以使模型更有效地提取互补特征，并实现最佳性能。相比之下，案例 2 和案例 3 中较早整合 FP 的情形，在所有数据集上表现较差。这表明过早融合 FP 会扰乱图像和图特征的结构化整合，从而导致表示性能不佳。案例 4 通过在连接之前分别处理所有模态对来取消融合顺序，在某些数据集（如 hERG 和 DILI）中表现具有竞争力。然而，总体上仍不及案例 1，凸显了结构化融合方法的优势。最后，案例 5 使用简单特征连接且未使用 MoE 跨模态注意力，在所有数据集上表现较低。这强调了显式建模模态间交互的重要性。总体而言，这些实验...

4.5. 消融研究

为了分析 MoE 对自注意力和交叉注意力模块的影响，我们进行了消融研究，比较了不同的配置。具体来说，我们考虑了三种类型的注意力：第一种是默认类型，即标准的多头注意力，不使用 MoE。第二种是使用密集 MoE 的多头注意力，其中所有专家都参与计算。最后一种是使用稀疏 MoE 的多头注意力，该方式会动态选择仅部分专家参与计算。通过评估这些配置的各种组合，我们旨在评估 MoE 在注意力机制中的应用。

图5展示了使用五个基准数据集在几种实验设置下的AUROC结果。结果表明，与默认注意力机制相比，将MoE与注意力机制结合通常能够提高性能，证明了基于专家的注意力机制在多模态学习中的优势。在这些配置中，同时将稀疏MoE应用于两个注意力模块的设置所有数据集上都取得了最高性能，凸显了专家选择在平衡计算效率和模型表现力方面的有效性。

在自注意力和交叉注意力模块中使用MoE被证明是有益的。对于密集MoE，在两个模块中都使用密集MoE的配置比只在一个模块中应用时的准确率更高，DILIst数据集除外。同样，稀疏MoE也表现出...

与密集 MoE 相同的趋势。值得注意的是，利用稀疏 MoE 不仅在准确性方面更有效，而且在计算成本上也表现更优（有关各基线计算效率的详细比较，请参见补充材料的图 S7）。此外，稀疏 MoE 可确保模型选择最相关的专家，从而优于其他情况。

5. 讨论

在这项研究中，所提出的模型MEMOL在多个药物毒性预测的基准数据集上表现出卓越的性能。具体来说，它在AUROC上最多比现有模型高出8.33%，在AUPRC上高出9.11%，展示了其强大的预测能力和泛化能力。这一显著的性能主要归功于模型的三个关键组件：（1）多模态学习，（2）基于MoE的注意力机制，以及（3）稀疏MoE。在本节中，我们将讨论这三个组件如何贡献于MEMOL的有效性。

MEMOL 的核心优势之一在于其多模态架构，该架构整合了异质分子表示，包括图像、图和分子指纹（FPs）。实验结果显示，与单独使用某一模态相比，同时使用三种模态在所有基准数据集上的性能均显著提升（见第 4.4.1 节）。这表明每种模态捕捉了分子特性的互补特征，通过它们的整合可以实现对分子毒性更加全面和准确的理解。具体而言，分子指纹反映化学和药理属性，而分子图像和图主要编码结构特性。整合这些多样化的视角使模型能够做出更准确的预测。

为了提高多模态学习的有效性，我们将 MoE 机制融合到多头注意力模块中。消融实验表明，与标准的多头注意力配置相比，应用 MoE 通常能够带来性能提升（详见第 4.5 节）。这表明，在注意力机制中利用专家选择能够实现输入特征更具适应性和专业化的处理。特别是，基于 MoE 的注意力结构能够捕捉复杂的模态特定特征，并在有效整合互补信息的同时减少跨模态重叠或冲突的风险。通过动态地将信息路由至专家子网络，模型实现了增强的表示能力。

此外，由于 MEMOL 使用的是稀疏 MoE 而非密集 MoE，该模型不仅在计算效率上具有优势，同时也具有较高的预测准确性。在密集 MoE 中，每个输入都会激活所有专家，这可能增加计算成本并在学习的表示中引入冗余。而稀疏 MoE 则只激活基于输入选择的一部分专家，从而实现更高效的计算，同时对性能的影响最小。在所有测试的配置中，将稀疏 MoE 应用于自注意力和交叉注意力块在基准数据集上获得了最高的 AUROC 分数。这些结果表明，选择性激活专家使模型能够更有效地关注相关模式，同时保持可控的计算负载。总体而言，对于多模态毒性预测任务——在这些任务中准确性和效率都至关重要——稀疏 MoE 似乎是更合适的选择。

6. 结论

可靠的药物毒性预测是药物开发中必不可少的组成部分，但由于分子机制的复杂性，这仍然是一项具有挑战性的任务。近年来，人工智能驱动的方法在解决这一问题方面显示出了巨大的潜力，尤其是整合各种分子数据表示的多模态方法。尽管如此

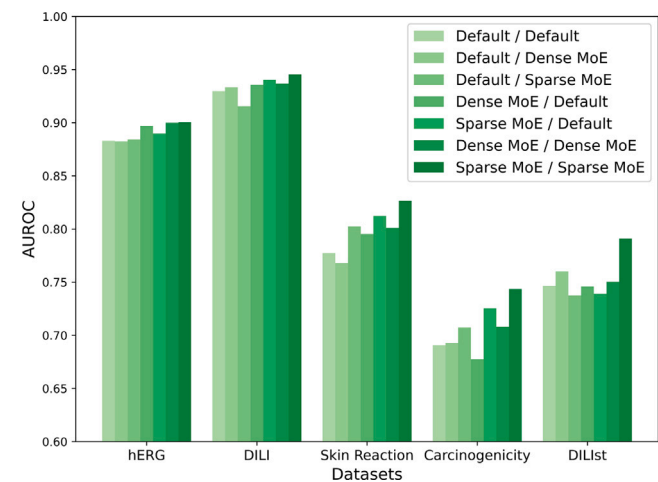


Fig. 5. AUROC results from the ablation study of the proposed model on five benchmark datasets. In the legend, the left part represents the method applied to self-attention, while the right part denotes the approach used for cross-attention.

graph first before incorporating FP, which was identified as the most informative modality in previous experiments, allows the model to extract complementary features more effectively and leads to optimal performance. In contrast, case 2 and case 3, where FP was integrated earlier, showed lower performance across all datasets. This indicates that fusing FP too early can disrupt the structured integration of image and graph features, resulting in suboptimal representations. Case 4, which removed fusion order by processing all modality pairs separately before concatenation, showed competitive results in certain datasets, such as hERG and DILI. However, it generally underperformed compared to case 1, highlighting the advantages of a structured fusion approach. Finally, case 5, which used simple feature concatenation without MoE cross-attention, showed low performance for all datasets. This highlights the importance of explicitly modeling inter-modality interactions. Overall, these experiments demonstrated not only the importance of usage of MoE cross-attention but also the order of modality integration.

4.5. Ablation study

To analyze the effect of MoE on self-attention and cross-attention blocks, we conducted an ablation study comparing different configurations. Specifically, we considered three types of attention: The first type refers to the default, which is a standard multi-head attention without MoE. The second type is multi-head attention using dense MoE where all experts contribute to computation. The last type is multi-head attention using sparse MoE which dynamically selects only a subset of experts. By evaluating various combinations of these configurations, we aim to assess the usage of MoE on attention mechanisms.

Fig. 5 presents the AUROC results using five benchmark datasets for several experimental settings. The results indicate that incorporating MoE with attention mechanisms generally improves performance compared to the default attention mechanism, demonstrating the benefits of expert-based attention mechanisms in multimodal learning. Among these configurations, the setting that applies sparse MoE to both attention blocks achieved the highest performance across all datasets, highlighting the effectiveness of expert selection in balancing computational efficiency and model expressiveness.

Utilizing MoE in both self- and cross-attention blocks proved beneficial. For dense MoE, the configuration using dense MoE for both blocks achieved higher accuracy than those applying it in only one block, except for the DILIst dataset. Similarly, sparse MoE exhibited the

same trend as dense MoE. Notably, leveraging sparse MoE was more effective not only in terms of accuracy but also in computational cost (see Figure S7 of the Supplementary Materials for detailed comparisons of computational efficiency across baselines). Furthermore, a sparse MoE ensures that the model selects the most relevant experts, which leads to outperforming other cases.

5. Discussion

In this study, the proposed model, MEMOL, achieved superior performance across multiple benchmark datasets for drug toxicity prediction. Specifically, it outperformed existing models by up to 8.33% in AUROC and 9.11% in AUPRC, demonstrating its strong predictive capability and generalizability. This remarkable performance can be primarily attributed to three key components of the model: (1) multimodal learning, (2) MoE-based attention, and (3) sparse MoE. In this section, we discuss how each of these components contributes to the effectiveness of MEMOL.

One of the core strengths of MEMOL lies in its multimodal architecture that integrates heterogeneous molecular representations, including images, graphs, and FPs. When compared to employing individual modalities alone, the experimental results show that the employment of all three modalities consistently resulted in considerable performance improvements across all benchmark datasets (see Section 4.4.1). This implies that each modality captures complementary features of molecular characteristics, and that a more thorough and accurate understanding of molecular toxicity is made possible by their integration. In particular, FPs reflect chemical and pharmacological attributes, whereas molecular images and graphs mostly encode structural properties. Integrating these diverse viewpoints allows the model to make more accurate predictions.

To improve the effectiveness of multimodal learning, we incorporated the MoE mechanism into multi-head attention blocks. The ablation study demonstrated that the application of MoE generally led to improved performance when compared to the standard multi-head attention configuration (see Section 4.5). This suggests that leveraging expert selection within the attention mechanism enables more adaptive and specialized processing of input features. In particular, the MoE-based attention structure is capable of capturing complex modality-specific characteristics and effectively integrating complementary information while reducing the risk of overlap or conflict across modalities. By dynamically routing information through expert sub-networks, the model achieves enhanced representational capacity.

Furthermore, since MEMOL utilizes sparse MoE instead of dense MoE, the model has advantages not only improved computational efficiency but also high predictive accuracy. In dense MoE, all experts are activated for each input, which can increase computational cost and introduce redundancy in the learned representations. On the other hand, sparse MoE activates only a subset of experts selected based on the input, leading to more efficient computation with minimal impact on performance. Among all tested configurations, applying sparse MoE to both self-attention and cross-attention blocks yielded the highest AUROC scores across the benchmark datasets. These results suggest that selective expert activation enables the model to focus on relevant patterns more effectively, while maintaining a manageable computational load. Overall, sparse MoE appears to be a more suitable choice for multimodal toxicity prediction tasks, where both accuracy and efficiency are crucial.

6. Conclusion

Reliable drug toxicity prediction is a necessary component of drug development, yet it remains a challenging task due to the complexity of molecular mechanisms. In recent years, AI-driven methods have shown great potential in addressing this issue, particularly multimodal approaches that integrate various molecular data representations. Despite

他们的成功表明，有效整合多种模态以提升毒性预测仍需进一步发展。为克服这些限制，我们提出了 MEMOL，一种基于 MoE 的多模态药物毒性预测框架，通过多头注意力机制实现。通过将稀疏 MoE 融入自注意力和交叉注意力机制，我们的模型有效优化了模态特定特征选择，同时促进了结构化的跨模态信息交换，最终提高了毒性预测的准确性。

在五个基准数据集上进行的实验表明，MEMOL 优于最先进的方法，在 AUROC 上比第二佳模型高出 8.33%，在 AUPRC 上高出 9.11%。特别是，配置为 $K = 2$ 和 $E = 4$ 的方案在各数据集上表现最为稳健，进一步证明了精心选择专家的重要性。模态组合实验的影响表明，利用所有三种模态对于捕捉全面的分子特征是必不可少的，同时也证明了 MEMOL 方法在优化整合顺序方面的有效性，即先融合图像和图表示，再整合 FP 表示。消融研究进一步验证了稀疏 MoE 在自注意力和交叉注意力机制中的有效性，凸显了为每个输入选择最相关专家的重要性。

未来的工作将探索在本研究中使用的模态上的自监督学习，以应对标注数据的缺乏，从而能够在更大规模的数据集上进行实验。此外，通过可解释的人工智能技术增强模型的可解释性，将有助于建立对人工智能驱动的毒性预测的信任。

CRedit 作者贡献声明

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附录 A. 补充数据

与本文相关的补充材料可在线查阅，网址为 <https://doi.org/10.1016/j.cmpb.2025.109088>。

代码和数据可用性

本研究中提出的源代码可在 <https://github.com/cjw0206/MEMOL> 获取。

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their success, effectively integrating multiple modalities to enhance toxicity prediction still requires further advancement. To overcome these limitations, we propose MEMOL, an MoE-based multimodal framework for drug toxicity prediction through multi-head attention. By incorporating sparse MoE into both self-attention and cross-attention mechanisms, our model effectively refines modality-specific feature selection while facilitating structured cross-modal information exchange, ultimately improving the accuracy of toxicity predictions.

Experiments conducted on five benchmark datasets demonstrated that MEMOL outperforms state-of-the-art methods, surpassing the second-best model by 8.33% in AUROC and 9.11% in AUPRC. In particular, the configuration with $K = 2$ and $E = 4$ yielded the most robust performance across datasets, reinforcing the benefits of careful expert selection. The impact of modality combinations experiments confirmed that utilizing all three modalities is essential for capturing comprehensive molecular characteristics, while also proving the effectiveness of the approach of MEMOL in optimizing the integration order by first fusing image and graph representations before integrating FP representations. The ablation study further validated the effectiveness of sparse MoE in both self-attention and cross-attention mechanisms, highlighting the importance of selecting the most relevant experts for each input.

Future work will explore self-supervised learning on the modalities used in this study to address the lack of labeled data, enabling experiments on larger-scale datasets. Additionally, enhancing model interpretability through explainable AI techniques will help build trust in AI-driven toxicity prediction.

CRedit authorship contribution statement

Jae-Woo Chu: Writing – original draft, Validation, Software, Data curation. Jong-Hoon Park: Writing – review & editing, Writing – original draft, Investigation, Formal analysis. Young-Rae Cho: Supervision, Investigation, Funding acquisition, Conceptualization.

Ethics statement

This research does not involve human participants and animal studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cmpb.2025.109088>.

Code and data availability

The source code proposed in this research is available at <https://github.com/cjw0206/MEMOL>.

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